DUMMY TENDER DOCUMENT

FINNED TYPE HEAT EXCHANGER FOR ROOM AIR CONDITIONER

Academic / Project Procurement Simulation

Tender Reference	DTHX/RAC/2026/001
Tender Type	Open / Dummy Technical & Commercial Tender
Issue Date	20 September 2026
Bid Validity	90 days from submission deadline
Procuring Entity	ABC HVAC Systems – Academic Simulation
Project	Design & Supply of Finned Type Heat Exchanger
Application	Room Air Conditioner

IMPORTANT: This document is a fictional tender prepared for academic/project demonstration only. It is not issued by any government department, PSU, company, or purchasing authority and has no contractual validity.

1. NOTICE INVITING TENDER

ABC HVAC Systems invites sealed technical and commercial offers from eligible manufacturers/suppliers for the design, manufacture, testing, supply and documentation of finned type heat exchangers intended for room air-conditioning applications.

Scope: Supply of evaporator/condenser-type finned heat exchanger coil assemblies suitable for integration into a room air conditioner, including tubes, fins, headers, return bends, mounting arrangements and associated quality documentation.

Submission: Bidders shall submit technical and commercial offers separately. Any deviations from the stated specifications shall be clearly identified.

2. TENDER INFORMATION

Item	Requirement
Tender Reference	DTHX/RAC/2026/001
Tender Document	Finned Type Heat Exchanger for Room Air Conditioner
Quantity	10 prototype units
Delivery Period	8 weeks from purchase order
Place of Delivery	ABC HVAC Systems – Dummy Project Site, India
Inspection	Pre-dispatch and/or receiving inspection
Warranty	12 months from commissioning or 18 months from dispatch, whichever occurs earlier

3. SCOPE OF SUPPLY

• Detailed thermal and mechanical design of the finned heat exchanger.

• Manufacture of coil assembly using approved tube and fin materials.

• Copper/aluminium tube-fin assembly with headers, return bends and end supports as applicable.

• Leak testing, dimensional inspection and performance verification.

• Protective packaging suitable for transportation and storage.

• Submission of drawings, material certificates, inspection reports and test certificates.

4. TECHNICAL SPECIFICATION

Parameter	Dummy Tender Requirement
Heat exchanger type	Finned-tube coil
Application	Room air conditioner
Configuration	Multi-row serpentine / parallel circuit coil
Tube material	Copper, suitable for refrigeration service
Fin material	Aluminium sheet, hydrophilic/coated option preferred
Tube outside diameter	Approximately 7–9.5 mm, final design by supplier
Fin thickness	Approximately 0.08–0.15 mm
Fin pitch	Approximately 1.5–2.5 mm
Rows	2–4 rows, subject to thermal design
Refrigerant	R-32 / equivalent as specified by purchaser
Design pressure	Minimum 4.2 MPa or as justified by applicable refrigerant/system design

Parameter	Dummy Tender Requirement
Air-side pressure drop	To be stated by bidder; target ≤ 50 Pa
Refrigerant-side pressure drop	To be stated by bidder
Operating temperature	Suitable for standard residential AC operating conditions
Corrosion protection	Suitable for indoor HVAC environment

5. DESIGN & PERFORMANCE REQUIREMENTS

• The supplier shall perform thermal design calculations based on the stated refrigerant, air-flow rate and required cooling capacity.

• The coil shall provide adequate heat transfer area while maintaining acceptable air-side and refrigerant-side pressure drops.

• The design shall consider fin efficiency, tube-side heat transfer, fouling allowance, condensate drainage and manufacturability.

• All refrigerant passages shall be clean, dry and free from contamination before sealing.

• The supplier shall provide the calculated heat-transfer capacity and design assumptions with the technical offer.

6. MATERIAL AND MANUFACTURING REQUIREMENTS

• Copper tubes shall be suitable for refrigeration applications and free from visible defects.

• Aluminium fins shall have uniform thickness and consistent fin pitch.

• Tube-to-fin contact shall be adequate to minimize thermal contact resistance.

• Brazed/soldered joints shall be clean, continuous and free from visible cracks or excessive filler material.

• Headers and return bends shall be compatible with the selected refrigerant and pressure rating.

• Sharp edges and burrs shall be removed from accessible surfaces.

• Manufacturing tolerances shall be stated on the approved manufacturing drawing.

7. INSPECTION AND TESTING

Test / Inspection	Acceptance Requirement
Visual inspection	No cracks, major dents, loose fins or unacceptable corrosion
Dimensional inspection	Conformance with approved drawing
Tube/coil leak test	No leakage at specified test pressure
Pressure test	As per approved design and applicable safety standard
Vacuum / cleanliness	Suitable for refrigeration-system assembly
Fin/tube bonding inspection	Uniform and mechanically secure
Performance test	Thermal capacity and pressure-drop results to be documented

8. QUALITY ASSURANCE

• Supplier shall maintain traceability of major materials used in the coil.

• Inspection records shall be retained and made available to the purchaser upon request.

• Any non-conforming component shall be identified and corrected/replaced before dispatch.

• The supplier shall submit a Certificate of Conformance with each batch.

9. DOCUMENTS TO BE SUBMITTED

• General arrangement drawing and detailed manufacturing drawing.

• Material certificates for tubes, fins and headers.

• Thermal design calculation and performance data.

• Pressure/leak test report.

• Inspection and dimensional report.

• Installation/handling instructions.

• Certificate of Conformance and warranty certificate.

10. BOQ / PRICE SCHEDULE

Sr.	Description	Qty.	Unit	Unit Price (■)	Total (■)
1	Finned type heat exchanger coil assembly	10	Nos.	_____	_____
2	Engineering/design documentation	1	Lot	_____	_____
3	Testing & inspection	1	Lot	_____	_____
4	Packaging & transportation	1	Lot	_____	_____
	Sub-total				_____
	GST / applicable taxes				_____
	Grand Total				_____

11. COMMERCIAL TERMS

• Prices shall include manufacturing, inspection, testing, standard packaging and all applicable costs unless separately stated.

• Taxes and duties shall be clearly indicated in the commercial offer.

• Payment terms: 30% advance, 60% after successful inspection, 10% after acceptance, as a dummy academic condition.

• Delivery shall be completed within the stated schedule.

• Any deviation from the commercial terms shall be explicitly stated.

12. WARRANTY

The supplier shall warrant the supplied heat exchanger against manufacturing defects and material defects for the specified warranty period. Defective components attributable to manufacturing or material defects shall be repaired or replaced at the supplier's cost, subject to the agreed warranty conditions.

13. BID SUBMISSION CHECKLIST

• Technical offer submitted.

• General arrangement / preliminary drawing submitted.

• Thermal design calculation submitted.

• Material details submitted.

• Testing methodology submitted.

• Delivery schedule submitted.

• Commercial offer / price schedule submitted.

• Deviation statement submitted.

• Warranty terms submitted.

14. VENDOR DECLARATION

We hereby declare that the information provided in our technical and commercial offer is accurate to the best of our knowledge and that the proposed heat exchanger can be manufactured and tested in accordance with the requirements stated in this dummy tender document.

Vendor Name	_____
Authorized Representative	_____
Designation	_____
Signature	_____
Date	_____

END OF DUMMY TENDER DOCUMENT

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